

NAME:
NEPTUN:

First Solution:
Second Solution:

1. Consider a 9-month futures contract on index. Suppose that the stocks underlying the index provide a dividend yield on 3% per annum (continuous compounded), that the current value of the index is 500. The futures price for a contract deliverable in four months is 525. What continuously compounded interest rate per annum do the spot and forward prices imply?

- a) 7,54%
- b) 9,54%
- c) 11,54%
- d) 12,54%

$$r = (\text{LN}(525/500)/0,75 + 0,03) = 9,54\%$$

2. Suppose that discount factors for different maturities (one two and three years) are given as 93%, 86% and 81%, respectively. The Treasury issued a 3-year maturity coupon bond, paying coupons once per year with a coupon rate of 6%. The face value of the bond is \$100. What is the arbitrage free 2-year forward price (delivery is right before the second year coupon payment)?

- a) \$98,84
- b) \$105,84
- c) \$108,84
- d) \$115,84

1	6	93%	5,58
2	6	86%	5,16
3	106	81%	85,86

Market price
96,6

$$\text{PV}(\text{Income during the forward contract}) = 6 * 0,93 = 5,58$$

$$F_2(0) = (96,6 - 5,58) / 0,86 = 105,84$$